

ESAT Physics Formula + Things to Remember

A compact, equation-first sheet for formulae, proportionalities, units, graph meanings and common physics traps.

ESAT Physics style: questions often reward unit sense, graph interpretation and proportional reasoning. Write the equation first, rearrange second, substitute last.

1. Units, graphs, motion and forces

ESAT SCOPE MARKERS

- core** Strictly Physics P1-P7, plus equation/unit shortcuts.
- trap** A-level fields, capacitors, circular motion, photoelectric effect and Newtonian gravity are intentionally removed.

UNITS AND PREFIXES

SI basics: m, kg, s, A, K. Derived units: N, J, W, C, V, Ω , Pa.
Useful prefixes: M = 10^6 , k = 10^3 , c = 10^{-2} , m = 10^{-3} , μ = 10^{-6} , n = 10^{-9} .
shortcut Keep powers of ten separate until the final step.

SCALARS AND VECTORS

Scalar = magnitude only. Vector = magnitude and direction.
Vectors: displacement, velocity, acceleration, force, momentum.
Scalars: distance, speed, mass, energy, time, temperature.

GRAPH INSTINCTS

gradient = $\Delta y / \Delta x$
Area under graph = product of axes.
v-t area = displacement; F-s area = work; I-t area = charge.

SPEED, VELOCITY, ACCELERATION

speed = distance / time
velocity = displacement / time
 $a = \Delta v / \Delta t$
Average speed uses total distance; average velocity uses displacement.

MOTION GRAPHS

- Distance-time gradient = speed.
- Displacement-time gradient = velocity.
- Velocity-time gradient = acceleration.
- Velocity-time area = displacement.
- Acceleration-time area = change in velocity.

CONSTANT ACCELERATION AND FREE FALL

$v^2 - u^2 = 2as$
Use only for constant acceleration.
Near Earth: $g \approx 10 \text{ N kg}^{-1} = 10 \text{ m s}^{-2}$.
shortcut No air resistance: horizontal velocity is constant; vertical acceleration is g.

FORCES, WEIGHT AND TERMINAL VELOCITY

$F = ma$; no resultant force $\rightarrow$ constant velocity.
 $W = mg$; weight is gravitational force.
Terminal velocity: drag/upthrust balances weight, resultant force = 0.

MOMENTUM AND NEWTON CHECKS

$p = mv$; momentum is conserved in a closed system.
force = rate of change of momentum.
Action-reaction forces are equal and opposite, same type, on different bodies.
trap Momentum and force are vectors; choose a positive direction.

2. Work, energy, springs, pressure and matter

WORK AND ENERGY TRANSFER

work done = force $\times$ distance moved in direction of force

$$W = Fs$$

Area under force-distance graph = work done.

GRAVITATIONAL POTENTIAL ENERGY

$$E_p = mgh$$

h is vertical height change.

Falling without losses: $mgh \rightarrow \frac{1}{2}mv^2$.

HOOKE LAW AND SPRINGS

$F = kx$ until limit of proportionality.

$$E = \frac{1}{2}Fx = \frac{1}{2}kx^2$$

Gradient of F - x graph = spring constant k ; area = elastic energy.

DENSITY

$$\rho = m / V$$

Units: kg m^{-3} or g cm^{-3} .

$$1 \text{ g cm}^{-3} = 1000 \text{ kg m}^{-3}$$

PARTICLE MODEL OF MATTER

- Solid: fixed shape/volume; particles vibrate in fixed positions.
- Liquid: fixed volume; particles close but move past each other.
- Gas: no fixed shape/volume; particles far apart and rapid.

SPECIFIC HEAT AND LATENT HEAT

$$E = mc\Delta T$$

$$E = mL$$

During melting/boiling, temperature does not rise while latent heat is supplied.

KINETIC ENERGY

$$E_k = \frac{1}{2}mv^2$$

shortcut Double speed $\rightarrow E_k \times 4$; double mass $\rightarrow E_k \times 2$.

POWER AND EFFICIENCY

$$P = E/t = W/t$$

efficiency = useful output / total input $\times 100\%$

Energy is conserved but can be dissipated to less useful stores.

ELASTIC VS INELASTIC

Elastic deformation: object returns to original shape when force removed.

Inelastic deformation: permanent change.

trap Hooke law only applies in proportional region.

PRESSURE

$$p = F/A$$

pressure in fluid = $h\rho g$

Fluid pressure increases with depth and density.

GASES AND STATE CHANGES

Gas pressure comes from particle collisions with container walls.

At constant temperature: $pV = \text{constant}$.

Change of state: temperature constant while energy changes particle separation/forces.

THERMAL TRANSFER

Conduction: particle/electron transfer; strongest in metals.

Convection: bulk fluid movement due to density differences.

Radiation: infrared EM waves; no medium needed.

3. Electrostatics, charge, circuits and resistance

ELECTROSTATICS

Insulators can be charged by friction through electron transfer.
Like charges repel; unlike charges attract.
Earthing allows charge to flow to/from Earth.

CHARGE AND CURRENT

$I = Q/t$
 $Q = It$
Conventional current is from positive to negative; electrons move opposite in metals.

POTENTIAL DIFFERENCE

$V = E/Q$
Voltage = energy transferred per unit charge.
 $1 \text{ V} = 1 \text{ J C}^{-1}$

RESISTANCE

$R = V/I$
 $V = IR$
Ohmic conductor: V-I graph straight through origin at constant temperature.

ELECTRICAL POWER AND ENERGY

$P = IV$
 $P = I^2R$
 $E = VIt = Pt$
trap Use E for energy, not electric field in ESAT circuits.

SERIES CIRCUITS

- Current same everywhere.
- Potential difference shared.
- Total resistance: $R_T = R_1 + R_2 + \dots$
- Adding series resistors increases total resistance.

PARALLEL CIRCUITS

- Potential difference same across each branch.
- Current splits at junctions.
- Total current = sum of branch currents.
- Total resistance is less than the smallest branch resistance.

CIRCUIT COMPONENTS

- Fixed resistor: constant resistance if temperature constant.
- Filament lamp: resistance rises as filament heats; V-I graph curves.
- NTC thermistor: resistance decreases as temperature increases.
- LDR: resistance decreases as light intensity increases.
- Diode: conducts mainly in one direction.

METERS AND PLACEMENT

Ammeter measures current; connect in series; very low resistance.
Voltmeter measures potential difference; connect in parallel; very high resistance.

AC/DC AND MATERIALS

Direct current flows one direction; alternating current reverses direction.
Conductors have mobile charge carriers; insulators have very few mobile charge carriers.

4. Magnetism, motor effect, induction and transformers

MAGNETS AND FIELDS

Like poles repel; unlike poles attract.

Magnetic field lines go from north to south outside a bar magnet.

Field direction = direction a north pole would move.

MAGNETIC MATERIALS

Soft magnetic materials magnetise/demagnetise easily; useful for electromagnets.

Hard magnetic materials retain magnetism; useful for permanent magnets.

Induced magnetism is usually attraction.

CURRENT MAGNETIC FIELDS

Current in a wire produces a circular magnetic field.

A coil/solenoid produces a field like a bar magnet.

Stronger field with larger current, more turns or iron core.

MOTOR EFFECT

A current-carrying wire in a magnetic field experiences a force.

$F = BIL$ for wire perpendicular to field.

Use Fleming left-hand rule for direction.

trap Force is zero if current is parallel to field.

DC MOTOR PRINCIPLE

Opposite forces on two sides of a coil create a turning moment.

Split-ring commutator reverses current every half-turn to keep rotation in same direction.

ELECTROMAGNETIC INDUCTION

A voltage is induced when a conductor cuts magnetic field lines or the magnetic field through a coil changes.

Magnitude increases with faster movement/change, stronger field, more turns or larger area.

LENZ DIRECTION IDEA

Induced current direction opposes the change that produced it.

shortcut If magnet is pushed into coil, induced field resists entry; if pulled out, it resists removal.

AC GENERATOR

Rotating coil in a magnetic field produces alternating voltage.

Output graph is alternating; maximum when rate of cutting field lines is greatest; zero when sides move parallel to field lines.

TRANSFORMERS

$$V_s/V_p = N_s/N_p$$

$$V_p I_p = V_s I_s \text{ for 100\% efficiency.}$$

Step-up: more secondary turns; higher voltage, lower current.

TRANSMISSION SHORTCUT

High voltage transmission lowers current for same power.

Lower current reduces heating losses in cables.

$P = IV$, so for fixed P : higher $V \rightarrow$ lower I .

5. Waves, sound, EM spectrum and radioactivity

WAVE BASICS

Waves transfer energy without net transfer of matter.
Transverse: oscillations perpendicular to energy transfer.
Longitudinal: oscillations parallel.
Sound is longitudinal; EM waves are transverse.

WAVE TERMS

- Amplitude: maximum displacement from rest position.
- Wavelength: distance between matching points.
- Frequency: cycles per second.
- Period: time for one cycle.
- Compression/rarefaction for longitudinal waves.

DOPPLER EFFECT

Observed frequency/wavelength changes when source and observer move relative to each other.
Approaching source: higher frequency, shorter wavelength.
Moving away: lower frequency, longer wavelength.

ULTRASOUND

Ultrasound frequency is above 20 kHz.
Uses include sonar and medical scanning; reflections at boundaries reveal distances/structures.

RADIOACTIVITY BASICS

Unstable nuclei decay randomly and spontaneously.
Decay is unaffected by temperature, pressure or chemical state.
Background radiation must be subtracted from measured count rate when needed.

ALPHA, BETA, GAMMA

Radiation	Nature	Penetration	Deflection
α	He nucleus	paper/skin	towards - plate
β^-	electron	thin aluminium	towards + plate
γ	EM wave	thick lead/concrete	none

HALF-LIFE AND FINAL TACTICS

Half-life = time for activity/count rate/undecayed nuclei to halve.

shortcut 3 half-lives $\rightarrow$ 1/8 remains; 4 $\rightarrow$ 1/16.

- Eliminate answers by units first.
- Check limiting cases: zero, doubled variable, very large value.
- For inverse-square-like wording inside spec, doubling distance usually quarters intensity.
- Direction: acceleration follows resultant force.

WAVE FORMULAE

$f = 1/T$
 $v = f\lambda$
 v in m s^{-1} , f in Hz, λ in m, T in s.

REFLECTION AND REFRACTION

Reflection: angle of incidence = angle of reflection.
Refraction occurs when wave speed changes at a boundary; frequency stays the same, wavelength changes.
trap Changing medium changes speed and wavelength, not frequency.

SOUND

Sound needs a medium and is produced by vibrating sources.
Loudness depends on amplitude; pitch depends on frequency.
Human hearing: about 20 Hz to 20 kHz. Echoes are reflections of sound.

EM SPECTRUM

radio $\rightarrow$ microwave $\rightarrow$ infrared $\rightarrow$ visible $\rightarrow$ ultraviolet $\rightarrow$ X-ray $\rightarrow$ gamma
Across this order: frequency and photon energy increase; wavelength decreases.
All EM waves travel at the same speed in vacuum.

NUCLEAR NOTATION AND DECAY

${}^A_Z\text{X}$: A = mass/nucleon number; Z = proton number; neutrons = A - Z.

Decay	A change	Z change
alpha	-4	-2
beta-minus	0	+1
gamma	0	0